How People Use ChatGPT*

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Abstract

Despite the rapid adoption of LLM chatbots, little is known about how they are used. We document the growth of ChatGPT's consumer product from its launch in November 2022 through July 2025, when it had been adopted by around 10% of the world's adult population. Early adopters were disproportionately male but the gender gap has narrowed dramatically, and we find higher growth rates in lower-income countries. Using a privacy-preserving automated pipeline, we classify usage patterns within a representative sample of ChatGPT conversations. We find steady growth in work-related messages but even faster growth in non-work-related messages, which have grown from 53% to more than 70% of all usage. Work usage is more common for educated users in highly-paid professional occupations. We classify messages by conversation topic and find that "Practical Guidance," "Seeking Information," and "Writing" are the three most common topics and collectively account for nearly 80% of all conversations. Writing dominates work-related tasks, highlighting chatbots' unique ability to generate digital outputs compared to traditional search engines. Computer programming and self-expression both represent relatively small shares of use. Overall, we find that ChatGPT provides economic value through decision support, which is especially important in knowledge-intensive jobs.

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1 Introduction

ChatGPT launched in November 2022. By July 2025, 18 billion messages were being sent each week by 700 million users, representing around 10% of the global adult population.[1] For a new technology, this speed of global diffusion has no precedent (Bick et al., 2024).

This paper studies consumer usage of ChatGPT, the first mass-market chatbot and likely the largest.[2] ChatGPT is based on a Large Language Model (LLM), a type of Artificial Intelligence (AI) developed over the last decade and generally considered to represent an acceleration in AI capabilities.[3]

The sudden growth in LLM abilities and adoption has intensified interest in the effects of artificial intelligence on economic growth (Acemoglu, 2024; Korinek and Suh, 2024); employment (Eloundou et al., 2025); and society (Kulveit et al., 2025). However, despite the rapid adoption of LLMs, there is limited public information on how they are used. A number of surveys have measured self-reported adoption of LLMs (Bick et al., 2024; Pew Research Center, 2025); however there are reasons to expect bias in self-reports (Ling and Imas, 2025), and none of these papers have been able to directly track the quantity or nature of chatbot conversations.

Two recent papers do report statistics on chatbot conversations, classified in a variety of ways (Handa et al., 2025; Tomlinson et al., 2025). We build on this work in several respects. First, the pool of users on ChatGPT is far larger, meaning we expect our data to be a closer approximation to the average chatbot user.[4] Second, we use automated classifiers to report on the types of messages that users send using new classification taxonomies relative to the existing literature. Third, we report the diffusion of chatbot use across populations and the growth of different types of usage within cohorts. Fourth, we use a secure data clean room protocol to analyze aggregated employment and education categories for a sample of our users, lending new insights about differences in the types of messages sent by different groups while protecting user privacy.

Our primary sample is a random selection of messages sent to ChatGPT on consumer plans (Free, Plus, Pro) between May 2024 and June 2025.[5] Messages from the user to chatbot are classified automatically using a number of different taxonomies: whether the message is used for paid work, the topic of conversation, and the type of interaction (asking, doing, or expressing), and the O*NET task the user is performing. Each taxonomy is defined in a prompt passed to an LLM, allowing us to classify messages without any human seeing them. We give the text of most prompts in Appendix A along with details about how the prompts were validated in Appendix B.[6] The classification pipeline is protected by a series of privacy measures, detailed below, to ensure no leakage of sensitive information during the automated analysis. In a secure data clean room, we relate taxonomies of messages to aggregated employment and education categories.

Table 1 shows the growth in total message volume for work and non-work usage. Both types of

[1] Reuters (2025), Roth (2025)

[2] Bick et al. (2024) report that 28% of US adults used ChatGPT in late 2024, higher than any other chatbot.

[3] We use the term LLM loosely here and give more details in the following section.

[4] Wiggers (2025) reports estimates that in April 2025 ChatGPT was receiving more than 10 times as many visitors as either Claude or Copilot.

[5] Our sample includes the three consumer plans (Free, Plus, or Pro). OpenAI also offers a variety of other ChatGPT plans (Business fka. Teams, Enterprise, Education), which we do not include in our sample.

[6] Our classifiers take into account not just the randomly-selected user message, but also a portion of the preceding messages in that conversation.

Month	Non-Work (M)	(%)	Work (M)	(%)	Total Messages (M)
Jun 2024	238	53%	213	47%	451
Jun 2025	1,911	73%	716	27%	2,627

Table 1: ChatGPT daily message counts (millions), broken down by likely work-related or non-work-related. Total daily counts are exact measurements of message volume from all consumer plans. Daily counts of work and non-work related messages are estimated by classifying a random sample of conversations from that day. Sampling is done to exclude users who opt-out of sharing their messages for model training, users who self-report their age as under 18, logged-out users, deleted conversations, and accounts which have been deactivated or banned (details available in Section 3). Reported values are 7-day averages (to smooth weekly fluctuation) ending on the 26th of June 2024 and 26th of June 2025.

messages have grown continuously, but non-work messages have grown faster and now represent more than 70% of all consumer ChatGPT messages. While most economic analysis of AI has focused on its impact on productivity in paid work, the impact on activity outside of work (home production) is on a similar scale and possibly larger. The decrease in the share of work-related messages is primarily due to changing usage within each cohort of users rather than a change in the composition of new ChatGPT users. This finding is consistent with Collis and Brynjolfsson (2025), who use choice experiments to uncover willingness-to-pay for generative AI and estimate a consumer surplus of at least $97 billion in 2024 alone in the US.

We next report on a classification of messages using a taxonomy developed at OpenAI for understanding product usage (\conversation classifier"). Nearly 80% of all ChatGPT usage falls into three broad categories, which we call *Practical Guidance*, *Seeking Information*, and *Writing*. *Practical Guidance* is the most common use case and includes activities like tutoring and teaching, how-to advice about a variety of topics, and creative ideation.[7] *Seeking Information* includes searching for information about people, current events, products, and recipes, and appears to be a very close substitute for web search. *Writing* includes the automated production of emails, documents and other communications, but also editing, critiquing, summarizing, and translating text provided by the user. *Writing* is the most common use case at work, accounting for 40% of work-related messages on average in June 2025. About two-thirds of all *Writing* messages ask ChatGPT to modify user text (editing, critiquing, translating, etc.) rather than creating new text from scratch. About 10% of all messages are requests for tutoring or teaching, suggesting that education is a key use case for ChatGPT.

Two of our findings stand in contrast to other work. First, we find the share of messages related to computer coding is relatively small: only 4.2% of ChatGPT messages are related to computer programming, compared to 33% of work-related Claude conversations Handa et al. (2025).[8] Second, we find the share of messages related to companionship or social-emotional issues is fairly small: only 1.9% of ChatGPT messages are on the topic of *Relationships and Personal Reflection* and 0.4% are related

[7] The difference between *Practical Guidance* and *Seeking Information* is that the former is highly customized to the user and can be adapted based on conversation and follow-up, whereas the latter is factual information that should be the same for all users. For example, users interested in running might ask ChatGPT for the Boston Marathon qualifying times by age and gender (*Seeking Information*), or they might ask for a customized workout plan that matches their goals and current level of fitness (*Practical Guidance*).

[8] Handa et al. (2025) report that 37% of conversations are mapped to a "computer and mathematical" occupation category, and their Figure 12 shows 30% or more of all imputed tasks are programming or IT-related. We believe the discrepancy is partly due to the difference in types of users between Claude and ChatGPT, additionally Handa et al. (2025) only includes queries that "possibly involve an occupational task".

to *Games and Role Play*. In contrast, Zao-Sanders (2025) estimates that *Therapy/Companionship* is the most prevalent use case for generative AI.[9]

We also document several important facts about demographic variation in ChatGPT usage. First, we show evidence that the gender gap in ChatGPT usage has likely narrowed considerably over time, and may have closed completely. In the few months after ChatGPT was released about 80% of active users had typically masculine first names.[10] However, that number declined to 48% as of June 2025, with active users slightly more likely to have typically feminine first names. Second, we find that nearly half of all messages sent by adults were sent by users under the age of 26, although age gaps have narrowed somewhat in recent months. Third, we find that ChatGPT usage has grown relatively faster in low- and middle-income countries over the last year. Fourth, we find that educated users and users in highly-paid professional occupations are substantially more likely to use ChatGPT for work.

We introduce a new taxonomy to classify messages according to the kind of output the user is seeking, using a simple rubric that we call *Asking, Doing,* or *Expressing.*[11] *Asking* is when the user is seeking information or clarification to inform a decision, corresponding to problem-solving models of knowledge work (e.g., Garicano (2000); Garicano and Rossi-Hansberg (2006); Carnehl and Schneider (2025); Ide and Talamas (2025)). *Doing* is when the user wants to produce some output or perform a particular task, corresponding to classic task-based models of work (e.g., Autor et al. (2003)). *Expressing* is when the user is expressing views or feelings but not seeking any information or action. We estimate that about 49% of messages are *Asking*, 40% are *Doing*, and 11% are *Expressing.* However, as of July 2025 about 56% of work-related messages are classified as *Doing* (e.g., performing job tasks), and nearly three-quarters of those are *Writing* tasks. The relative frequency of writing-related conversations is notable for two reasons. First, writing is a task that is common to nearly all white-collar jobs, and good written communication skills are among the top "soft" skills demanded by employers (National Association of Colleges and Employers, 2024). Second, one distinctive feature of generative AI, relative to other information technologies, is its ability to produce long-form outputs such as writing and software code.

We also map message content to work activities using the Occupational Information Network (O*NET), a survey of job characteristics supported by the U.S. Department of Labor. We find that about 58% of work-related messages are associated with two broad work activities: 1) obtaining, documenting, and interpreting information; and 2) making decisions, giving advice, solving problems, and thinking creatively. Additionally, we find that the work activities associated with ChatGPT usage are highly similar across very different kinds of occupations. For example, the work activities *Getting Information* and *Making Decisions and Solving Problems* are in the top five of message frequency in nearly all occupations, ranging from management and business to STEM to administrative and sales occupations.

Overall, we find that information-seeking and decision support are the most common ChatGPT use cases in most jobs. This is consistent with the fact that almost half of all ChatGPT usage is either *Practical Guidance* or *Seeking Information*. We also show that *Asking* is growing faster than

[9] Zao-Sanders (2025) is based on a manual collection and labeling of online resources (Reddit, Quora, online articles), and so we believe it likely resulted in an unrepresentative distribution of use cases.

[10] Among those with names commonly associated with a particular gender.

[11] Appendix A gives the full prompt text and Appendix B gives detail about how the prompts were validated against public conversation data.

Doing, and that Asking messages are consistently rated as having higher quality both by a classi er that measures user satisfaction and from direct user feedback.

How does ChatGPT provide economic value, and for whom is its value the greatest? We argue that ChatGPT likely improves worker output by providing decision support, which is especially important in knowledge-intensive jobs where better decision-making increases productivity (Deming, 2021; Caplin et al., 2023). This explains why Asking is relatively more common for educated users who are employed in highly-paid, professional occupations. Our ndings are most consistent with Ide and Talamas (2025), who develop a model where AI agents can serve either as co-workers that produce output or as co-pilots that give advice and improve the productivity of human problem-solving.

2 What is ChatGPT?

Here we give a simpli ed overview of LLMs and chatbots. For more precise details, refer to the papers and system cards that OpenAI has released with each model e.g., (OpenAI, 2023, 2024a, 2025b). A chatbot is a statistical model trained to generate a text response given some text input, so as to maximize the \quality" of that response, where the quality is measured with a variety of metrics.

In a prototypical interaction, a user submits a plain-text message (\prompt") and ChatGPT returns the message (\response") generated from an underlying LLM. A large set of additional features have been added to ChatGPT|including the possibility for the LLM to search the web or external databases, and generate images based on text|but the exchange of text-based messages remains the most typical interaction.

Since its launch ChatGPT has used a variety of di erent underlying LLMs e.g., GPT-3.5, GPT-4, GPT-4o, o1, o3, and GPT-5.[12] In addition there are occasional updates to the model's weights and to the model's system prompt (text instructions sent to the model along with all the queries).

An LLM can be thought of as a function from a string of words to a probability distribution over the set of all possible words (more precisely, \tokens," which very roughly correspond to words[13]). The functions are implemented with deep neural nets, typically with a transformer architecture (Vaswani et al., 2017), parameterized with billions of model \weights". We will refer to all of ChatGPT's models as language models, though most can additionally process tokens representing images, audio, or other media.

The weights in an LLM-based chatbot are often trained in two stages, commonly called \pre-training" and \post-training". In the rst stage (\pre-training"), the LLMs are trained to predict the next word in a string, given the preceding words, over an enormous corpus of text. At that point the models are purely predictors of the likelihood of the next word given a prior context, and as such they have a relatively narrow application. In the second stage (\post-training"), the models are trained to produce words that comprise \good" responses to some prompt. This stage often consists of a variety of di erent strategies: ne-tuning on a dataset of queries and ideal responses, reinforcement learning against another model that is trained to grade the quality of a response (Ouyang et al., 2022), or reinforcement learning against a function that knows the true response to queries (OpenAI (2024b),

[12] For a timeline of model launches, see Appendix C.

[13] Tokenization is a way of cutting a string of text into discrete chunks, chosen to be statistically e cient. In many tokenization schemes, one token corresponds to roughly three-quarters of an English word.

Lambert et al. (2024)). This second stage also typically includes a number of \safety" constraints to avoid certain classes of response, especially those which are deemed harmful or dangerous (OpenAI, 2025a).

This two-stage process has a common statistical interpretation: the rst stage teaches the model a latent representation of the world; the second stage ts a function using that representation (Bengio et al., 2014). Pre-training the model to predict the next word e ectively teaches the model a low-dimensional representation of text, representing only the key semantic features, and therefore rendering the prompt-response problem tractable with a reasonable set of training examples.

Two common ways of evaluating chatbots are with benchmarks (batteries of questions with known answers, e.g. Measuring Massive Multitask Language Understanding (Hendrycks et al., 2021)) and comparisons of human preferences over two alternative responses to the same message (e.g. Chatbot Arena (Chiang et al., 2024)).

3 Data and Privacy

In this section, we describe the data used in the paper and the privacy safeguards we implemented. No member of the research team ever saw the content of user messages, and all analyses were conducted in accordance with OpenAI's Privacy Policy (OpenAI, 2025c).

The analysis in this paper is based on the following datasets:

1. Growth: total daily message volumes from consumer ChatGPT users between November 2022 and September 2025, along with basic self-reported demographic information. This dataset is primarily used in Section 4.
2. Classi ed messages: messages classi ed into coarse categories.
   - Sampled from all ChatGPT users: a random sample of approximately one million de-identi ed messages from logged-in consumer ChatGPT users between May 2024 and June 2025.[14] This dataset is primarily used in Section 5.
   - Sampled from a subset of ChatGPT users: two random samples of messages sent between May 2024 and July 2025 by a subset of consumer ChatGPT users (one sample at the conversation level, one sample at the user level).[15] These datasets are primarily used in Section 6.
3. Employment: aggregated employment and education categories based on publicly available data for a subset of consumer ChatGPT users. This data is only used in Section 6.

We describe the contents of each dataset, the sampling procedures that produced them, and the privacy protections we implemented in constructing and employing them in analysis.

3.1 Growth Dataset

We compiled a dataset covering all usage on consumer ChatGPT Plans (Free, Plus, Pro) since Chat-GPT's launch in November 2022. We exclude users on non-consumer plans (Business f.k.a. Teams,

[14] The exact beginning and end dates of this sample are May 15, 2024 and June 26, 2025.
[15] The exact beginning and end dates of this sample are May 15, 2024 and July 31, 2025.

Enterprise, Education).

For each user and day, this dataset reports the total number of messages sent by the user on that day. It also reports, for each message, de-identified user metadata, including the timestamp of their first interaction with ChatGPT, the country from which their account is registered, their subscription plan on each day, and their self-reported age (reported in coarse 5–7-year buckets to protect user privacy).

3.2 Classified Messages

To understand usage while preserving user privacy, we construct message-level datasets without any human ever reading the contents of a message. See Figure 1 for an overview of the privacy-preserving classification pipeline. Messages are categorized according to 5 different LLM-based classifiers. The classifiers are introduced in more detail in Section 5, their exact text is reproduced in Appendix A, and our validation procedure is described in Appendix B.

Sampled From All ChatGPT Users. We uniformly sampled approximately 1.1 million conversations, and then sampled one message within each conversation, with the following restrictions:

1. We only include messages from May 2024 to July 2025.
2. We exclude conversations from users who opted out of sharing their messages for model training.
3. We exclude users who self-report their age as under 18.
4. We exclude conversations that users have deleted and from users whose accounts have been deactivated or banned.
5. We exclude logged-out users,[16] which represented a minority share of ChatGPT users over the sample period.

Our sample is drawn from a table that is itself sampled, where the sampling rate varied over time. We thus adjust our sampling weights to maintain a fixed ratio with aggregate messages sent.

Sampled From a Subset of ChatGPT Users. We construct two samples of classified messages from a subset of ChatGPT users (approximately 130,000 users). This sample of users does not include any users who opted out of sharing their messages for training, nor does it include users whose self-reported age is below 18, nor does it include users who have been banned or deleted their accounts.

The first sample contains classifications of 1.58 million messages from this subset of users, sampled at the conversation level (a conversation is a series of messages between the user and chatbot). This sample is constructed such that the user's representation in the data is proportional to overall message volume. The second sample contains messages sent from this subset of users, sampled at the user level with up to six messages from each user in the group.

[16] ChatGPT became available to logged-out users in April 2024, i.e., users could use ChatGPT without signing up for an account with an email address. However, messages from logged-out users are only available in our dataset from March 2025, thus for consistency we drop all messages from logged-out users.

Figure 1: Illustration of Privacy-Preserving Automated Classi cation Pipeline (Synthetic Example). Messages are rst stripped of PII via an internal LLM-based tool called Privacy Filter. Then they are classi ed by LLM-based automated classi ers, described in detail in Appendices A and B. Humans do not see raw messages or PII-scrubbed messages, only the nal classi cations of messages.

Privacy via Automated Classi ers. No one looked at the content of messages while conducting analysis for this paper. All analysis of message content was performed via automated LLM-based classi ers run on de-identi ed and PII-scrubbed message data (see Figure 1). The messages are rst scrubbed of PII using an internal LLM-based tool,[17] and then classi ed according to classi ers de ned over a controlled label space|the most precise classi er we use on the message-level data set is the O*NET Intermediate Work Activities taxonomy, which we augment to end up with 333 categories. We introduce technical and procedural frictions that prevent accidental access to the underlying text (for example, interfaces that do not render message text to researchers).

Our classi cations aim to discern the intent of a given message, and thus we include the prior 10 messages in a conversation as context.[18] For an example, see Table 2.

Stand-Alone Message	Message with Prior Context
[user]: \10 more"	[user]: \give me 3 cultural activities to do with teens" [assistant]: \1. Visit a museum . . . " (truncated) [user]: \10 more"

Table 2: Illustration of Context-Augmented Message Classi cations (Synthetic Example). The left column shows a standalone message to be classi ed, and the right column shows the prior context included in the classi cation of the message on the left.

We truncate each message to a maximum of 5,000 characters, because long context windows could induce variability in the quality of the classi cation (Liu et al., 2023). We classify each message with the \gpt-5-mini" model, with the exception of Interaction Quality, which uses \gpt-5," using the prompts listed in Appendix A.

[17] Internal analyses show that the tool, Privacy Filter, has substantial alignment with human judgment.
[18] In the case of Interaction Quality, we additionally include the next two messages in the conversation as context.

We validated each of the classi cation prompts by comparing model classi cation decisions against human-judged classi cations of a sample of conversations from the publicly available WildChat dataset (Zhao et al., 2024), a set of conversations with a third-party chatbot which users a rmatively gave their assent to share publicly for research purposes.[19] Appendix B provides detail on our validation approach and performance relative to human judgment. For additional transparency, we classify a sample of 100,000 public WildChat messages and provide those data in this paper's replication package.

3.3 Employment Dataset

We conduct limited analyses of aggregated employment categories based on publicly available data for a sample of consumer ChatGPT users. This sample included approximately 130,000 Free, Plus, and Pro users, and the employment categories were aggregated by a vendor working through a secure Data Clean Room (DCR). For this analysis, we use the same exclusion criteria as for the message-level datasets: we exclude deactivated users, banned users, users who have opted out of training, and users whose self-reported age is under 18. Because the data was only available for a subset of users the results may not be representative of the full pool of users.

Description. The employment data, which is aggregated from publicly available sources, includes industry, occupations coarsened to O*NET categories, seniority level, company size, and education information that is limited to the degree attained. A vendor working within a DCR procured this dataset, restricted us to running only aggregated queries against it through the DCR, and deleted it upon the study's completion.

Privacy via a Data Clean Room. We never directly accessed user-level demographic records. All analysis of employment data was executed exclusively within a secure DCR that permits only pre-approved aggregate computations across independently held datasets; neither party can view or export the other party's underlying records. We governed the DCR with strict protocols: To execute any query that touched the external demographic data, we rst obtained explicit sign-o from a committee of 6 coauthors and then submitted the notebook to our data partner for approval; only approved notebooks could run in the DCR (see Figure 2).

Our partner enforced strict aggregation limits: they only approved code that returned cells meeting a threshold of 100 users. Consequently, no individual rows or narrowly de ned categories were ever visible to researchers. For example, if 99 users had the occupation \anesthesiologist," any occupation-level output would place those users into a \suppressed" category, or place these observations in a coarsened category (e.g. \medical professionals") rather than reporting a separate cell of anesthesiologists.

[19] The dataset was collected from a third party chatbot using OpenAI's LLMs via their API.

Figure 2: Illustration of Aggregated Employment Category Analysis via a Data Clean Room. All queries run in the Data Clean Room must be approved by our data partner, enforcing a strict aggregation threshold (100 observations). As a result, researchers cannot access user-level employment data, only aggregated employment categories.

3.4 Summarizing Our Approach to Privacy

We took measures to safeguard user privacy at every stage of analysis. To summarize, the key elements of our approach are:

> Automated classi cation of messages. In the course of analysis, no one ever looked directly at the content of user messages: all of our analysis of the content of user messages is done through output of automated classi ers run on de-identi ed and PII-scrubbed usage data.
>
> Aggregated employment data via a data clean room. We analyze and report aggregated employment data through a secure data clean room environment: no one on the research team had direct access to user-level demographic data and none of our analyses report aggregates for groups with less than 100 users.

In following these measures, we aim to match or exceed the privacy protection precedents set by other social scientists studying chatbots and those linking digital platform data to external sources.

We follow the precedent established in recent analyses of chatbot conversations (Phang et al. (2025), Eloundou et al. (2025), Handa et al. (2025), Tomlinson et al. (2025)) that rely on automated classi cation rather than human inspection of raw transcripts. In particular, Phang et al. (2025)'s study of a ective use of ChatGPT and Eloundou et al. (2025) investigation of rst-person fairness in chatbots both analyze ChatGPT message content via automated classi ers and emphasize classi er-based labeling as a scalable, privacy-preserving approach. Anthropic's Handa et al. (2025) used a similar approach: their Clio methodology applies automated classi ers to large collections of conversations, classifying conversations into thousands of topics, and in their appendix they describe manual validation on sampled conversations (100 user conversations agged for review and 100 randomly sampled calibrations). Like Eloundou et al., we validate our classi ers using WildChat, a public dataset of user conversations.

Other papers have analyzed digital behavior and demographic data; we mention a few relevant precedents here. Humlum and Vestergaard (2025b) and Humlum and Vestergaard (2025a), for example, analyze large-scale surveys on chatbot use along with Danish administrative labor market data. Chetty et al. (2022) analyze de-identi ed Facebook friendship graphs and anonymized IRS tax records, aggregated at the zip code level.

4 The Growth of ChatGPT

ChatGPT was released to the public on November 30, 2022 as a \research preview," and by December 5 it had more than one million registered users. Figure 3 reports the growth of overall weekly active users (WAU) on consumer plans over time. ChatGPT had more than 100 million logged-in WAU after one year, and almost 350 million after two years. By the end of July 2025, ChatGPT had more than 700 million total WAU, nearly 10% of the world's adult population. [20]

Figure 3: Weekly active ChatGPT users on consumer plans (Free, Plus, Pro), shown as point-in-time snapshots every six months, November 2022{September 2025.

Figure 4 presents growth in the total messages sent by users over time. The solid line shows that between July 2024 and July 2025, the number of messages sent grew by a factor of more than 5.

Figure 4 also shows the contribution of individual cohorts of users to aggregate message volume. The yellow line represents the rst cohort of ChatGPT users: their usage declined somewhat over 2023, but started growing again in late 2024 and is now higher than it has ever been. The pink line represents messages from users who signed up in Q3 of 2023 or earlier, and so the di erence between

[20] Note that we expect our counts of distinct accounts to somewhat exceed distinct people when one person has two accounts (or, for logged-out users, one person using two devices). For logged-in users, the count is based on distinct login credentials (email addresses), and one person may have multiple accounts. For logged-out users, the count is based on distinct browser cookies; this would double-count people if someone returns to ChatGPT after clearing their cookies, or if they access ChatGPT with two di erent devices in the same week.

Figure 4: Daily message volumes from ChatGPT consumer plans (Free, Plus, Pro), split by sign-up date of the requesting user. Reported values are moving averages of the past 90 days. Y-axis is an index normalized to the reported value for "All Cohorts" at the end of Q1 2024 (April 1, 2024).

the yellow and pink lines represents the messages sent by users who signed up in Q2 and Q3 of 2023. There has been dramatic growth in message volume both by new cohorts of users, and from growth in existing cohorts.

Figure 5 normalizes each cohort, plotting daily messages per weekly active user. Each line represents an individual cohort (instead of a cumulative cohort, as in Figure 4). The gure shows that earlier sign-ups have consistently had higher usage, but that usage has also consistently grown within every cohort, which we interpret as due to both (1) improvements in the capabilities of the models, and (2) users slowly discovering new uses for existing capabilities.

5 How ChatGPT is Used

We next report on the content of ChatGPT conversations using a variety of di erent taxonomies. For each taxonomy we describe a \prompt" which de nes a set of categories, and then apply an LLM to map each message to a category. Our categories often apply to the user's intention, rather than the text of the conversation, and as such we never directly observe the ground truth. Nevertheless the classi er results can be interpreted as the best-guess inferences that a human would make: the guesses from the LLM correlate highly with human guesses from the same prompt, and we get similar qualitative results when the prompt includes a third category for \uncertain."

Figure 5: Daily messages sent per weekly active user, split by sign-up cohort. Sample only considers users of ChatGPT consumer plans (Free, Plus, Pro). Reported values are moving averages of the past 90 days and are reported starting 90 days after the cohort is fully formed. Y-axis is an index normalized to the rst reported value for the Q1 2023 cohort.

5.1 What share of ChatGPT queries are related to paid work?

We label each user message in our dataset based on whether it appears to be related to work, using an LLM classi er. The critical part of the prompt is as follows: [21]

> Does the last user message of this conversation transcript seem likely to be related to doing some work/employment? Answer with one of the following:
>
> (1) likely part of work (e.g., \rewrite this HR complaint")
>
> (0) likely not part of work (e.g., \does ice reduce pimples?")

Table 1 shows that both types of queries grew rapidly between June 2024 and June 2025, however non-work-related messages grew faster: 53% of messages were not related to work in June 2024, which climbed to 73% by June 2025.

Figure 6 plots the share of non-work messages decomposed by cumulative sign-up cohorts. Successive cohorts have had a higher share of non-work messages, but also within each cohort their non-work use has increased. Comparing the share among all users (black line) to the share among the earliest cohort of users (yellow line), we can see that they track very closely.

[21] See Appendix A for the full prompt, see Appendix B for validation.

Figure 6: The solid black line represents the probability that a messages on a given day is not related to work, as determined by an automated classi er. Values are averaged over a 28-day lagging window. The dotted orange line shows the same calculation, but conditioned on messages being from users who rst used ChatGPT during or before Q2 of 2024. The remaining lines are de ned similarly for successive quarters, with coloring cooling for more recent cohorts. Counts are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to re ect total message volumes on a given day. Sampling details available in Section 3.

5.2 What are the topics of ChatGPT conversations?

We modify a classi er used by internal research teams at OpenAI that identi es which capabilities the user is requesting from ChatGPT. The classi er itself directly assigns the user's query into one of 24 categories. We aggregate these 24 categories into seven topical groupings (the full conversation-categorization prompt is given in Appendix A):

Topic	Conversation Category
Writing	Edit or Critique Provided Text
	Personal Writing or Communication
	Translation
	Argument or Summary Generation
	Write Fiction
Practical Guidance	How-To Advice
	Tutoring or Teaching
	Creative Ideation
	Health, Fitness, Beauty, or Self-Care
Technical Help	Mathematical Calculation
	Data Analysis

Topic	Conversation Category
	Computer Programming
Multimedia	Create an Image
	Analyze an Image
	Generate or Retrieve Other Media
Seeking Information	Speci c Info
	Purchasable Products
	Cooking and Recipes
Self-Expression	Greetings and Chitchat
	Relationships and Personal Re ection
	Games and Role Play
Other/Unknown	Asking About the Model
	Other
	Unclear

Table 3: Coarse Conversation Topics and Underlying Classi er Categories

Figure 7 shows the composition of user messages over time. The three most common Conversation Topics are Practical Guidance, Seeking Information, and Writing, collectively accounting for about 77% of all ChatGPT conversations. Practical Guidance has remained constant at roughly 29% of overall usage. Writing has declined from 36% of all usage in July 2024 to 24% a year later. Seeking Information has grown from 14% to 24% of all usage over the same period. The share of Technical Help declined from 12% from all usage in July 2024 to around 5% a year later { this may be because the use of LLMs for programming has grown very rapidly through the API (outside of ChatGPT), for AI assistance in code editing and for autonomous programming agents (e.g. Codex). Multimedia grew from 2% to just over 7%, with a large spike in April 2025 after ChatGPT released new image-generation capabilities: the spike attenuated but the elevated level has persisted.

Figure 8 shows Conversation Topics, restricting the sample to only work-related messages. About 40% of all work-related messages in July 2025 are Writing, by far the most common Conversation Topic. Practical Guidance is the second most common use case at 24%. Technical Help has declined from 18% of all work-related messages in July 2024 to just over 10% in July 2025.

Figure 9 disaggregates four of the seven Conversation Topics into smaller groups and sums up messages of each type over a one-year period. For example, the ve sub-categories within Writing are (in order of frequency) Editing or Critiquing Provided Text, Personal Writing or Communication, Translation, Argument or Summary Generation, and Writing Fiction. Three of those ve categories (Editing or Critiquing Provided Text, Translation, and Argument or Summary Generation) are requests to modify text that has been provided to ChatGPT by the user, whereas the other two are requests to produce novel text. The former constitute two thirds of all Writing conversations, which

Figure 7: Share of consumer ChatGPT messages broken down by high level conversation topic, according to the mapping in Table 3. Values are averaged over a 28 day lagging window. Shares are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to re ect total message volumes on a given day. Sampling details available in Section 3.

Figure 8: Share of work related consumer ChatGPT messages broken down by high level conversation topic, according to the mapping in Table 3. Values are averaged over a 28 day lagging window. Shares are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to re ect total message volumes on a given day. Sampling details available in Section 3.

suggests that most user Writing conversations with ChatGPT are requests to modify user inputs rather than to create something new. Education is a major use case for ChatGPT. 10.2% of all user messages and 36% of Practical Guidance messages are requests for Tutoring or Teaching. Another large share - 8.5% in total and 30% of Practical Guidance - is general how-to advice on a variety of topics. Technical Help includes Computer Programming (4.2% of messages), Mathematical Calculations (3%), and Data Analysis (0.4%). Looking at the topic of Self-Expression, only 2.4% of all ChatGPT messages are about Relationships and Personal Re ection (1.9%) or Games and Role Play (0.4%).

While users can seek information and advice from traditional web search engines as well as from ChatGPT, the ability to produce writing, software code, spreadsheets, and other digital products distinguishes generative AI from existing technologies. ChatGPT is also more exible than web search even for traditional applications like Seeking Information and Practical Guidance, because users receive customized responses (e.g., tailored workout plans, new product ideas, ideas for fantasy football team names) that represent newly generated content or novel modi cation of user-provided content and follow-up requests.

Figure 9: Breakdown of granular conversation topic shares within the coarse mapping de ned in Table 3. The underlying classi er prompt is available in Appendix A. Each bin reports a percentage of the total population. Shares are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to re ect total message volumes on a given day. Sampling details available in Section 3.

5.3 User Intent

Existing studies of the economic impacts of generative AI focus almost exclusively on the potential for AI to perform workplace tasks, either augmenting or automating human labor (e.g. Eloundou et al. (2025), Handa et al. (2025), Tomlinson et al. (2025)). However, generative AI is a highly exible

technology that can be used in many di erent ways. In order to learn more about how people seek to use generative AI at work and outside of work, we introduce a classi er that is designed to measure the type of output the user hopes to receive. Speci cally, we classify messages according to user intent, coding up conversations according to a simple Asking, Doing, or Expressing rubric. The critical part of our classi cation prompt is as follows:

Intent	Prompt
Asking	Asking is seeking information or advice that will help the user be better informed or make better decisions, either at work, at school, or in their personal life. (e.g. \Who was president after Lincoln?", \How do I create a budget for this quarter?", \What was the in ation rate last year?", \What's the di erence between correlation and causation?", \What should I look for when choosing a health plan during open enrollment?").
Doing	Doing messages request that ChatGPT perform tasks for the user. User is drafting an email, writing code, etc. Classify messages as \doing" if they include requests for output that is created primarily by the model. (e.g. \Rewrite this email to make it more formal", \Draft a report summarizing the use cases of ChatGPT", \Produce a project timeline with milestones and risks in a table", \Extract companies, people, and dates from this text into CSV.", \Write a Docker le and a minimal docker-compose.yml for this app.")
Expressing	Expressing statements are neither asking for information, nor for the chatbot to perform a task.

Conceptually, Doing conversations are delivering output that can be plugged into a production process, while Asking conversations support decision-making but do not produce output directly, and Expressing conversations have little or no economic content.

Figure 10 shows the share of messages by each intent type in our sample. 49% of user messages are Asking, 40% are Doing, and 11% are Expressing. The gure also shows the relationship with our Topic classi cation: the two taxonomies are correlated but not redundant: Asking queries are more likely to be Practical Guidance and Seeking Information. Doing queries are disproportionately Writing and Multimedia. Expressing queries are disproportionately Self-Expression. However, the overlap is imperfect. For example, within the Practical Guidance topic, an Asking message might be advice about how to recover from a sports injury given a user's personal history, while a Doing message might request ChatGPT to produce a customized recovery and training plan that could be printed or saved. Within Technical Help, an Asking message might request help understanding how to debug some code, while a Doing message might ask ChatGPT to write code for the user directly.

Figure 11 presents shares of Asking/Doing/Expressing just for work-related messages. Doing constitutes nearly 56% of work-related queries, compared to 35% for Asking and 9% for Expressing. Nearly 35% of all work-related queries are Doing messages related to Writing. Doing and Asking comprise equal shares of Technical Help queries.

Figure 10: Breakdown of Conversation Topics by Asking/Doing/Expressing category, with topic columns sorted by relative share of "Doing" messages. Prompts for these automated classi ers are available in Appendix A. For a detailed breakdown of conversation topic contents, see Table 3. Each bin reports a percentage of the total population. Shares are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to re ect total message volumes on a given day. Sampling details available in Section 3.

Figure 11: Breakdown of Conversation Topics by Asking/Doing/Expressing category for only work-related messages, with topic columns sorted by relative share of "Doing" messages. Prompts for these automated classi ers are available in Appendix A. For a detailed breakdown of conversation topic contents, see Table 3. Each bin reports a percentage of the total population. Shares are calculated from a sample of approximately 1.1 million sampled conversations from May 15, 2024 through June 26, 2025. Observations are reweighted to re ect total message volumes on a given day. Sampling details available in Section 3.